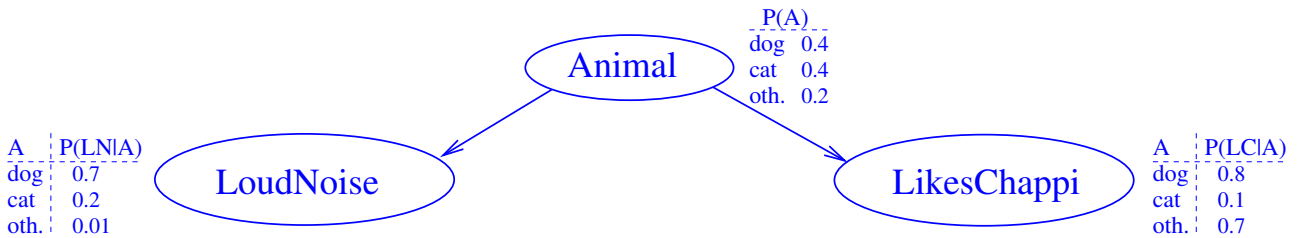


**Exercise 1 :**

Consider the following Bayesian network  $BN$ :



Use inference by enumeration to compute the following probabilities:

- $P(\text{dog} \mid \text{loudnoise}, \text{likeschappi})$ .
- $P(\text{loudnoise} \mid \neg \text{likeschappi})$ .

Include intermediate steps at a level of granularity as in the lecture slides examples. In particular, for each of (a) and (b), state what the query variable, evidence, and hidden variables are; and write down *which* probabilities provided in  $BN$  can be combined *how* to obtain the demanded probability  $P$ .

**Exercise 2 :**

In the graphs in Figures 1 and 2, determine which variables are d-separated from  $A$ . Note that it is sufficient to find a single open path along which evidence can be transmitted; if such a path exists then the variables are *not* d-separated (instead they are said to be d-connected).

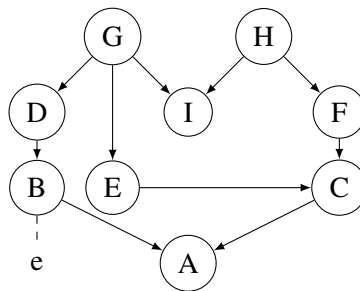


Figure 1: Figure for Exercise 2.

**Exercise 3 :**

Consider the network in Figure 3.

- What are the minimal set(s) of variables that we should have evidence on in order to d-separate  $C$  and  $E$  (that is, sets of variables for which no proper subset d-separates  $C$  and  $E$ )?
- What are the minimal set(s) of variables we should have evidence on in order to d-separate  $A$  and  $B$ ?
- What are the maximal set(s) of variables that we can have evidence on and still d-separate  $C$  and  $E$  (that is, sets of variables for which no proper superset d-separates  $C$  and  $E$ )?

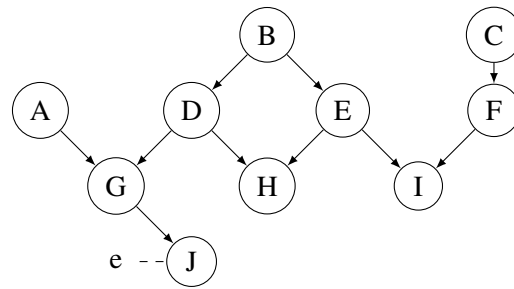


Figure 2: Figure for Exercise 2.

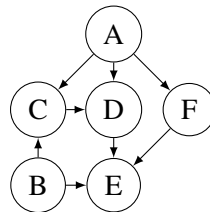


Figure 3: A causal network for Exercise ??.

- What are the maximal set(s) of variables that we can have evidence on and still d-separate  $A$  and  $B$ ?

**Exercise 4 :**

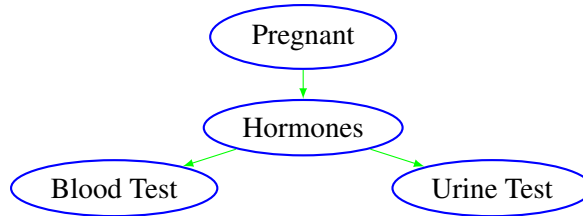
Consider the network defined by the two binary variables  $A$  and  $B$ , where  $A$  is the parent of  $B$ . Assume that the conditional probability tables are given as  $P(A) = (0.1, 0.9)$  and

|       | $A$   |       |
|-------|-------|-------|
|       | $a_1$ | $a_2$ |
| $b_1$ | 0.05  | 0.2   |
| $b_2$ | 0.95  | 0.8   |

- Assume that you want to estimate  $P(b_1)$  using sampling. How many samples would be required if you only accept an error larger than 0.15 in 10% of the cases?
- Generate two random samples, using the following list of random numbers (any time that you use a random number pick one from the list): 0.5, 0.3, 0.01, 0.8.
- Suppose that after sampling 10 states, you got  $(A = a_2, B = b_2)$  3 times, and  $(A = a_1, B = b_2)$  6 times, and  $(A = a_1, B = b_1)$  once. Estimate  $P(B = b_1)$
- (Optional) Implement the network above in Hugin and use Hugin to sample the number of cases that you calculated in step (i); use the function 'Simulate cases' under 'File'. Use the sampled cases to estimate  $P(b_1)$  and compare the result with Hugin. Feel free to use a spreadsheet for the counting.
- Assume that you want to use rejection sampling to estimate  $P(A|B = b_1)$ . How many samples do you expect you would have to generate in order to end up (after rejection) with a sample set of 1000 cases for estimating the probability.

**Exercise 5 :**

Consider the insemination example from Section 3.1.13 in BNDG:



Let the probabilities be as in Table 1 ( $Ho = y$  means that hormonal changes have taken place)  $P(Pr) = (0.87, 0.13)$ .

|          | $Pr = y$ | $Pr = n$ |          | $Ho = y$ | $Ho = n$ |
|----------|----------|----------|----------|----------|----------|
| $Ho = y$ | 0.9      | 0.01     | $BT = y$ | 0.7      | 0.1      |
| $Ho = n$ | 0.1      | 0.99     | $BT = n$ | 0.3      | 0.9      |

|          | $Ho = y$ | $Ho = n$ |
|----------|----------|----------|
| $UT = y$ | 0.8      | 0.1      |
| $UT = n$ | 0.2      | 0.9      |

Table 1: Tables for Exercise 5.

- What is  $P(Pr | BT = n, UT = n)$ ?
- (Optional) Verify your solution modelling the BN in Hugin.

**Exercise 6 :**

Complete Exercise 9.4(a-b) in PM.

**Exercise 7 :**

Consider the running example of the lecture with the following probability tables:

|  |          |     |  |            |     |
|--|----------|-----|--|------------|-----|
|  | Burglary |     |  | Earthquake |     |
|  | $t$      | $f$ |  | $t$        | $f$ |
|  | .1       | .9  |  | .1         | .9  |

|          |            | Alarm |     |
|----------|------------|-------|-----|
| Burglary | Earthquake | $t$   | $f$ |
| $t$      | $t$        | .9    | .1  |
| $t$      | $f$        | .8    | .2  |
| $f$      | $t$        | .5    | .5  |
| $f$      | $f$        | .1    | .9  |

| JohnCalls |     |     | MaryCalls |     |     |
|-----------|-----|-----|-----------|-----|-----|
| Alarm     | $t$ | $f$ | Alarm     | $t$ | $f$ |
| $t$       | .8  | .2  | $t$       | .7  | .3  |
| $f$       | .1  | .9  | $f$       | .1  | .9  |

Determine the conditional probability  $P(MC \mid B = t)$ . For each step indicate the operations that are performed.

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